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Branch	CMPN/B
Problem Statement	6

Screenshots

```
> # Install packages (run only once if not already installed)
> install.packages(c("palmerpenguins", "ggplot2", "dplyr", "moments", "car", "effsize"))

also installing the dependencies 'colorspace', 'fracdiff', 'lme4', 'lmtest', 'timeDate', 'urca', 'zoo', 'R
cppArmadillo', 'cowplot', 'Deriv', 'forecast', 'rbibutils', 'numDeriv', 'doBy', 'SparseM', 'Matrix
Models', 'Rdpack', 'minqa', 'nloptr', 'reformulas', 'RcppEigen', 'carData', 'Formula', 'pbkrtest',
'quantreg', 'lme4'

trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/colorspace_2.1-3.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/fracdiff_1.5-4.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/lmtest_0.9-40.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/timeDate_4052.112.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/urca_1.3-4.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/zoo_1.9-0.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/RcppArmadillo_15.4.2-1.tg
z'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/cowplot_1.2.0.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/Deriv_4.3.5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/forecast_9.0.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/rbibutils_2.4.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/numDeriv_2016.8-1.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/doBy_4.7.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/SparseM_1.84-2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/MatrixModels_0.5-4.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/Rdpack_2.6.6.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/minqa_1.2.8.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/nloptr_2.2.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/reformulas_0.4.4.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/RcppEigen_0.3.4.0.2.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/carData_3.0-6.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/Formula_1.2-6.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/pbkrtest_0.5.5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/quantreg_6.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/lme4_2.0-6.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/palmerpenguins_0.1.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/ggplot2_4.0.3.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/dplyr_1.2.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/moments_0.14.1.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/car_3.1-5.tgz'
trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/effsize_0.8.1.tgz'

The downloaded binary packages are in
  /var/folders/gs/wdrg4sjd3jq4mjrpdhcgvr0000gn/T//RtmpbaaFN9/downloaded_packages
>
> # Load packages
> library(palmerpenguins)
```

```
> library(ggplot2)
> library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
> library(moments)
> library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

```
> library(effsize)
> # Load Palmer Penguins dataset
> data("penguins")
>
> # View first few rows
> head(penguins)
```

A tibble: 6 × 8

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
	<fct>	<fct>	<dbl>	<dbl>	<int>	<int>	<fct>	<int>
1	Adelie	Torgersen	39.1	18.7	181	3750	male	2007
2	Adelie	Torgersen	39.5	17.4	186	3800	female	2007
3	Adelie	Torgersen	40.3	18	195	3250	female	2007
4	Adelie	Torgersen	NA	NA	NA	NA	NA	2007
5	Adelie	Torgersen	36.7	19.3	193	3450	female	2007
6	Adelie	Torgersen	39.3	20.6	190	3650	male	2007

```
>
```

```

> # Structure of dataset
> str(penguins)
tibble [344 × 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...

>
> # Summary
> summary(penguins)
      species      island  bill_length_mm  bill_depth_mm  flipper_length_mm
Adelie :152  Biscoe :168   Min. :32.10   Min. :13.10   Min. :172.0
Chinstrap: 68  Dream  :124   1st Qu.:39.23   1st Qu.:15.60   1st Qu.:190.0
Gentoo   :124  Torgersen: 52   Median :44.45   Median :17.30   Median :197.0
                                     Mean  :43.92   Mean  :17.15   Mean  :200.9
                                     3rd Qu.:48.50   3rd Qu.:18.70   3rd Qu.:213.0
                                     Max.  :59.60   Max.  :21.50   Max.  :231.0
                                     NAs   :2       NAs   :2       NAs   :2
 body_mass_g      sex      year
Min. :2700  female:165   Min. :2007
1st Qu.:3550  male :168   1st Qu.:2007
Median :4050  NAs  : 11   Median :2008
Mean :4202
3rd Qu.:4750
Max. :6300
NAs :2

>
> # Number of observations
> nrow(penguins)
[1] 344

>
> # Species
> unique(penguins$species)
[1] Adelie  Gentoo  Chinstrap
Levels: Adelie Chinstrap Gentoo

>

```

```

> colSums(is.na(penguins))
  species      island bill_length_mm bill_depth_mm flipper_length_mm
      0           0           2           2           2
body_mass_g      sex      year
      2          11           0
> # Dataset for body mass analysis
> penguin_data <- penguins %>%
+   filter(!is.na(body_mass_g),
+         !is.na(species),
+         !is.na(sex))
>
> # View cleaned data
> head(penguin_data)
# A tibble: 6 x 8
  species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g sex      year
  <fct>   <fct>      <dbl>      <dbl>      <int>      <int> <fct>  <int>
1 Adelie  Torgersen    39.1      18.7      181      3750 male    2007
2 Adelie  Torgersen    39.5      17.4      186      3800 female 2007
3 Adelie  Torgersen    40.3      18       195      3250 female 2007
4 Adelie  Torgersen    36.7      19.3      193      3450 female 2007
5 Adelie  Torgersen    39.3      20.6      190      3650 male    2007
6 Adelie  Torgersen    38.9      17.8      181      3625 female 2007
>
> # Number of observations after cleaning
> nrow(penguin_data)
[1] 333
> body_mass <- penguin_data$body_mass_g
>
> mean(body_mass)
[1] 4207.057
> median(body_mass)
[1] 4050
> min(body_mass)
[1] 2700
> max(body_mass)
[1] 6300
> var(body_mass)
[1] 648372.5
> sd(body_mass)
[1] 805.2158
> quantile(body_mass, 0.25)
25%
3550
> quantile(body_mass, 0.75)
75%
4775

```

```

> IQR(body_mass)
[1] 1225
> skewness(body_mass)
[1] 0.4701162
> kurtosis(body_mass)
[1] 2.259514
> species_stats <- penguin_data %>%
+   group_by(species) %>%
+   summarise(
+     Mean = mean(body_mass_g),
+     Median = median(body_mass_g),
+     Minimum = min(body_mass_g),
+     Maximum = max(body_mass_g),
+     Variance = var(body_mass_g),
+     SD = sd(body_mass_g),
+     Q1 = quantile(body_mass_g, 0.25),
+     Q3 = quantile(body_mass_g, 0.75),
+     IQR = IQR(body_mass_g),
+     Skewness = skewness(body_mass_g),
+     Kurtosis = kurtosis(body_mass_g)
+   )
>
> print(species_stats)
# A tibble: 3 x 12
  species      Mean Median Minimum Maximum Variance      SD      Q1      Q3      IQR Skewness Kurtosis
  <fct>      <dbl>  <dbl>   <int>   <int>      <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl>  <dbl>
1 Adelie    3706.   3700    2850    4775  210332.  459.  3362.  4000   638.    0.277    2.41
2 Chinstrap 3733.   3700    2700    4800  147713.  384.  3488.  3950   462.    0.242    3.46
3 Gentoo    5092.   5050    3950    6300  251478.  501.  4700   5500   800    0.0450    2.24

```

```

> ggplot(penguin_data, aes(x = body_mass_g)) +
+   geom_histogram(bins = 30) +
+   labs(
+     title = "Histogram of Penguin Body Mass",
+     x = "Body Mass (g)",
+     y = "Frequency"
+   ) +
+   theme_minimal()
> ggplot(penguin_data, aes(x = species, y = body_mass_g)) +
+   geom_boxplot() +
+   labs(
+     title = "Body Mass by Penguin Species",
+     x = "Species",
+     y = "Body Mass (g)"
+   ) +
+   theme_minimal()
> ggplot(penguin_data, aes(x = sex, y = body_mass_g)) +
+   geom_boxplot() +
+   labs(
+     title = "Body Mass by Sex",
+     x = "Sex",
+     y = "Body Mass (g)"
+   ) +
+   theme_minimal()
> ggplot(penguin_data, aes(x = body_mass_g)) +
+   geom_density() +
+   labs(
+     title = "Density Plot of Penguin Body Mass",
+     x = "Body Mass (g)",
+     y = "Density"
+   ) +
+   theme_minimal()
> # Male normality
> shapiro.test(
+   penguin_data$body_mass_g[penguin_data$sex == "male"]
+ )

      Shapiro-Wilk normality test

data:  penguin_data$body_mass_g[penguin_data$sex == "male"]
W = 0.92504, p-value = 1.227e-07

>

```

```

> # Female normality
> shapiro.test(
+   penguin_data$body_mass_g[penguin_data$sex == "female"]
+ )

      Shapiro-Wilk normality test

data:  penguin_data$body_mass_g[penguin_data$sex == "female"]
W = 0.91931, p-value = 6.155e-08

> # Male QQ plot
> qqnorm(
+   penguin_data$body_mass_g[penguin_data$sex == "male"],
+   main = "QQ Plot - Male Body Mass"
+ )
> qqline(
+   penguin_data$body_mass_g[penguin_data$sex == "male"]
+ )
>
> # Female QQ plot
> qqnorm(
+   penguin_data$body_mass_g[penguin_data$sex == "female"],
+   main = "QQ Plot - Female Body Mass"
+ )
> qqline(
+   penguin_data$body_mass_g[penguin_data$sex == "female"]
+ )
> t_test_result <- t.test(
+   body_mass_g ~ sex,
+   data = penguin_data,
+   var.equal = FALSE
+ )
>
> print(t_test_result)

```

Welch Two Sample t-test

```

data:  body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -840.5783 -526.2453
sample estimates:
mean in group female   mean in group male
      3862.273           4545.685

> cohen_result <- cohen.d(
+   penguin_data$body_mass_g[penguin_data$sex == "male"],
+   penguin_data$body_mass_g[penguin_data$sex == "female"]
+ )
>
> print(cohen_result)

Cohen's d

d estimate: 0.9362048 (large)
95 percent confidence interval:
      lower      upper
0.7090938 1.1633158

```

```
> ggplot(penguin_data, aes(x = species, y = body_mass_g)) +  
+   geom_boxplot() +  
+   labs(  
+     title = "Body Mass Distribution Across Species",  
+     x = "Species",  
+     y = "Body Mass (g)"  
+   ) +  
+   theme_minimal()  
> by(  
+   penguin_data$body_mass_g,  
+   penguin_data$species,  
+   shapiro.test  
+ )  
penguin_data$species: Adelie
```

Shapiro-Wilk normality test

```
data: dd[x, ]  
W = 0.98116, p-value = 0.04232
```

penguin_data\$species: Chinstrap

Shapiro-Wilk normality test

```
data: dd[x, ]  
W = 0.98449, p-value = 0.5605
```

penguin_data\$species: Gentoo

Shapiro-Wilk normality test

```
data: dd[x, ]  
W = 0.98606, p-value = 0.2605
```

```

> anova_model <- aov(
+   body_mass_g ~ species,
+   data = penguin_data
+ )
>
> summary(anova_model)
              Df    Sum Sq Mean Sq F value Pr(>F)
species        2 145190219 72595110   341.9 <2e-16 ***
Residuals    330  70069447   212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> leveneTest(
+   body_mass_g ~ species,
+   data = penguin_data
+ )
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group  2  5.1349 0.006367 **
      330
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> tukey_result <- TukeyHSD(anova_model)
>
> print(tukey_result)
      Tukey multiple comparisons of means
      95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = penguin_data)

$species
              diff            lwr            upr    p adj
Chinstrap-Adelie  26.92385 -132.3528  186.2005 0.916431
Gentoo-Adelie    1386.27259 1252.2897 1520.2554 0.000000
Gentoo-Chinstrap 1359.34874 1194.4304 1524.2671 0.000000

```

```

> 
> plot(tukey_result)
> kruskal_result <- kruskal.test(
+   body_mass_g ~ species,
+   data = penguin_data
+ )
> 
> print(kruskal_result)

Kruskal-Wallis rank sum test

data: body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

> # ANOVA
> summary(anova_model)
      Df    Sum Sq Mean Sq F value Pr(>F)    
species    2 145190219  72595110   341.9 <2e-16 ***
Residuals 330  70069447   212332                
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> 
> # Kruskal-Wallis
> kruskal_result

Kruskal-Wallis rank sum test

data: body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

> two_way_model <- aov(
+   body_mass_g ~ species * sex,
+   data = penguin_data
+ )
> 
> summary(two_way_model)
      Df    Sum Sq Mean Sq F value    Pr(>F)    
species    2 145190219  72595110  758.358 < 2e-16 ***
sex        1  37090262  37090262  387.460 < 2e-16 ***
species:sex  2  1676557    838278   8.757 0.000197 ***
Residuals 327  31302628    95727                
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

> interaction.plot(
+   penguin_data$species,
+   penguin_data$sex,
+   penguin_data$body_mass_g,
+   xlab = "Species",
+   ylab = "Mean Body Mass (g)",
+   trace.label = "Sex",
+   main = "Interaction Plot: Species and Sex"
+ )
> ggplot(
+   penguin_data,
+   aes(x = species, y = body_mass_g, fill = sex)
+ ) +
+   geom_boxplot() +
+   labs(
+     title = "Body Mass by Species and Sex",
+     x = "Species",
+     y = "Body Mass (g)",
+     fill = "Sex"
+   ) +
+   theme_minimal()
> flipper_data <- penguins %>%
+   filter(
+     !is.na(flipper_length_mm),
+     !is.na(species)
+   )
>
> # Descriptive statistics
> flipper_stats <- flipper_data %>%
+   group_by(species) %>%
+   summarise(
+     Mean = mean(flipper_length_mm),
+     Median = median(flipper_length_mm),
+     SD = sd(flipper_length_mm),
+     Minimum = min(flipper_length_mm),
+     Maximum = max(flipper_length_mm)
+   )
>
> print(flipper_stats)
# A tibble: 3 × 6
  species    Mean Median    SD Minimum Maximum
  <fct>    <dbl>  <dbl> <dbl>   <int>   <int>
1 Adelie   190.    190  6.54    172    210
2 Chinstrap 196.    196  7.13    178    212
3 Gentoo   217.    216  6.48    203    231

```

```

> ggplot(
+   flipper_data,
+   aes(x = species, y = flipper_length_mm)
+ ) +
+   geom_boxplot() +
+   labs(
+     title = "Flipper Length Across Penguin Species",
+     x = "Species",
+     y = "Flipper Length (mm)"
+   ) +
+   theme_minimal()
> flipper_anova <- aov(
+   flipper_length_mm ~ species,
+   data = flipper_data
+ )
>
> summary(flipper_anova)
              Df Sum Sq Mean Sq F value Pr(>F)
species         2  52473   26237    594.8 <2e-16 ***
Residuals      339  14953         44
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> flipper_tukey <- TukeyHSD(flipper_anova)
>
> print(flipper_tukey)
Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = flipper_length_mm ~ species, data = flipper_data)

$species
      diff      lwr      upr p adj
Chinstrap-Adelie  5.869887  3.586583  8.153191    0
Gentoo-Adelie    27.233349 25.334376 29.132323    0
Gentoo-Chinstrap 21.363462 19.000841 23.726084    0
>

```

```

> plot(flipper_tukey)
> flipper_kruskal <- kruskal.test(
+   flipper_length_mm ~ species,
+   data = flipper_data
+ )
>
> print(flipper_kruskal)

      Kruskal-Wallis rank sum test

data:  flipper_length_mm by species
Kruskal-Wallis chi-squared = 244.89, df = 2, p-value < 2.2e-16

> ggplot(
+   penguin_data,
+   aes(x = species, y = body_mass_g)
+ ) +
+   geom_boxplot() +
+   stat_summary(
+     fun = mean,
+     geom = "point",
+     shape = 18,
+     size = 3
+   ) +
+   labs(
+     title = "Penguin Body Mass Comparison Across Species",
+     x = "Penguin Species",
+     y = "Body Mass (g)"
+   ) +
+   theme_minimal()
> cat("\n===== FINAL ANALYSIS =====\n")

===== FINAL ANALYSIS =====
>
> cat("\n1. DESCRIPTIVE STATISTICS\n")

1. DESCRIPTIVE STATISTICS
> print(species_stats)
# A tibble: 3 x 12
  species    Mean Median Minimum Maximum Variance    SD    Q1    Q3    IQR Skewness Kurtosis
  <fct>    <dbl>  <dbl>   <int>   <int>    <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl>   <dbl>
1 Adelie  3706.   3700    2850    4775  210332.  459. 3362. 4000  638.   0.277    2.41
2 Chinstrap 3733.   3700    2700    4800  147713.  384. 3488. 3950  462.   0.242    3.46
3 Gentoo   5092.   5050    3950    6300  251478.  501. 4700  5500  800.   0.0450    2.24

```

```

> cat("\n2. MALE vs FEMALE T-TEST\n")

2. MALE vs FEMALE T-TEST
> print(t_test_result)

Welch Two Sample t-test

data: body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and group male is not equal
to 0
95 percent confidence interval:
-840.5783 -526.2453
sample estimates:
mean in group female mean in group male
3862.273 4545.685

>
> cat("\n3. COHEN'S D\n")

3. COHEN'S D
> print(cohen_result)

Cohen's d

d estimate: 0.9362048 (large)
95 percent confidence interval:
lower upper
0.7090938 1.1633158

>
> cat("\n4. ONE-WAY ANOVA\n")

4. ONE-WAY ANOVA
> print(summary(anova_model))
      Df    Sum Sq Mean Sq F value Pr(>F)
species    2 145190219 72595110  341.9 <2e-16 ***
Residuals 330 70069447  212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> cat("\n5. TUKEY HSD\n")

5. TUKEY HSD

```

```

> print(tukey_result)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = penguin_data)

$species
      diff      lwr      upr    p adj
Chinstrap-Adelie  26.92385 -132.3528  186.2005 0.916431
Gentoo-Adelie    1386.27259 1252.2897 1520.2554 0.000000
Gentoo-Chinstrap 1359.34874 1194.4304 1524.2671 0.000000

>
> cat("\n6. KRUSKAL-WALLIS TEST\n")

6. KRUSKAL-WALLIS TEST
> print(kruskal_result)

      Kruskal-Wallis rank sum test

data:  body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

>
> cat("\n7. TWO-WAY ANOVA\n")

7. TWO-WAY ANOVA
> print(summary(two_way_model))
      Df Sum Sq Mean Sq F value    Pr(>F)
species    2 145190219 72595110 758.358 < 2e-16 ***
sex         1  37090262 37090262  387.460 < 2e-16 ***
species:sex  2   1676557   838278    8.757 0.000197 ***
Residuals 327  31302628   95727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> cat("\n8. FLIPPER LENGTH ANOVA\n")

8. FLIPPER LENGTH ANOVA
> print(summary(flipper_anova))
      Df Sum Sq Mean Sq F value    Pr(>F)
species    2   52473   26237   594.8 <2e-16 ***
Residuals 339  14953     44
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>

```

```

> cat("\n9. FLIPPER LENGTH KRUSKAL-WALLIS\n")

9. FLIPPER LENGTH KRUSKAL-WALLIS
> print(flipper_kruskal)

      Kruskal-Wallis rank sum test

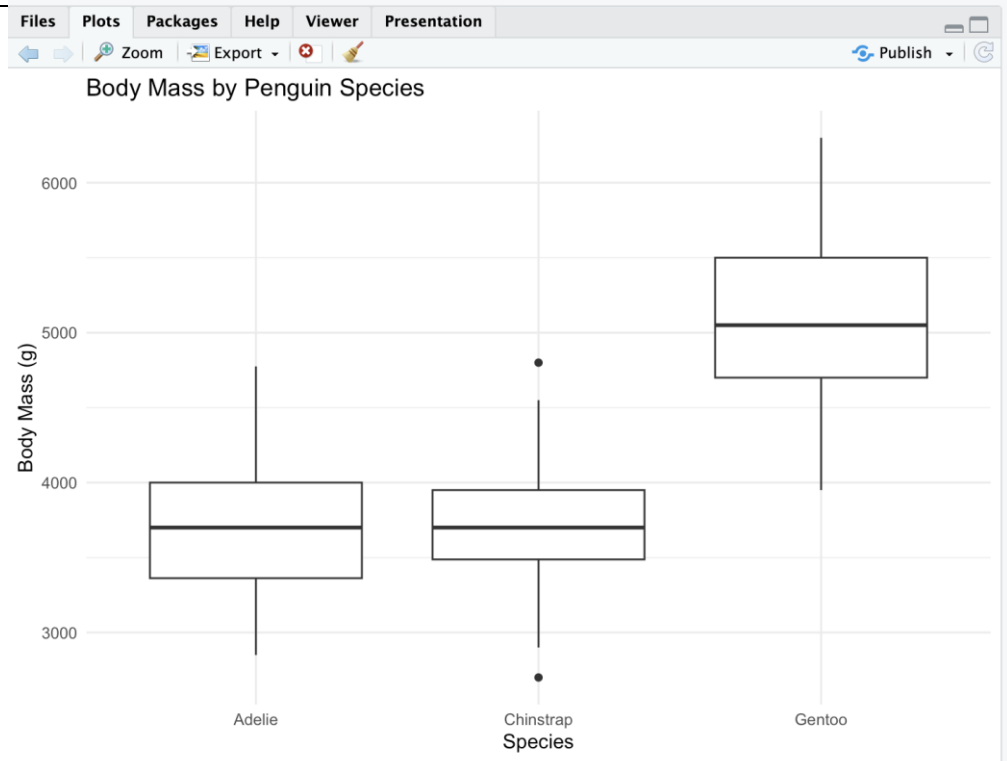
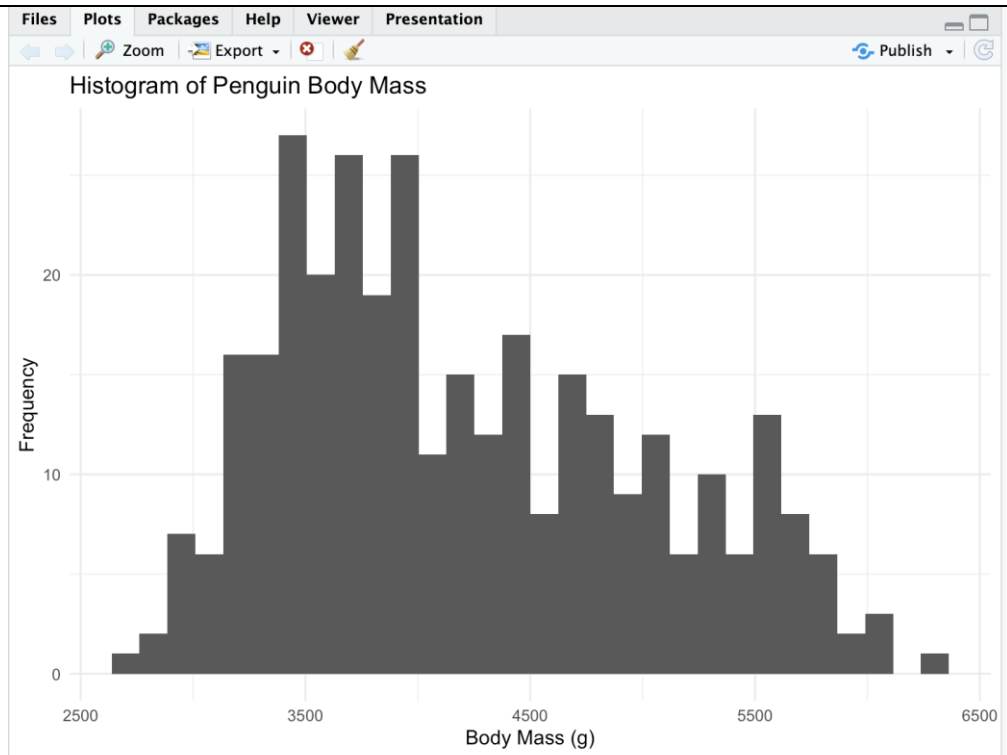
data:  flipper_length_mm by species
Kruskal-Wallis chi-squared = 244.89, df = 2, p-value < 2.2e-16

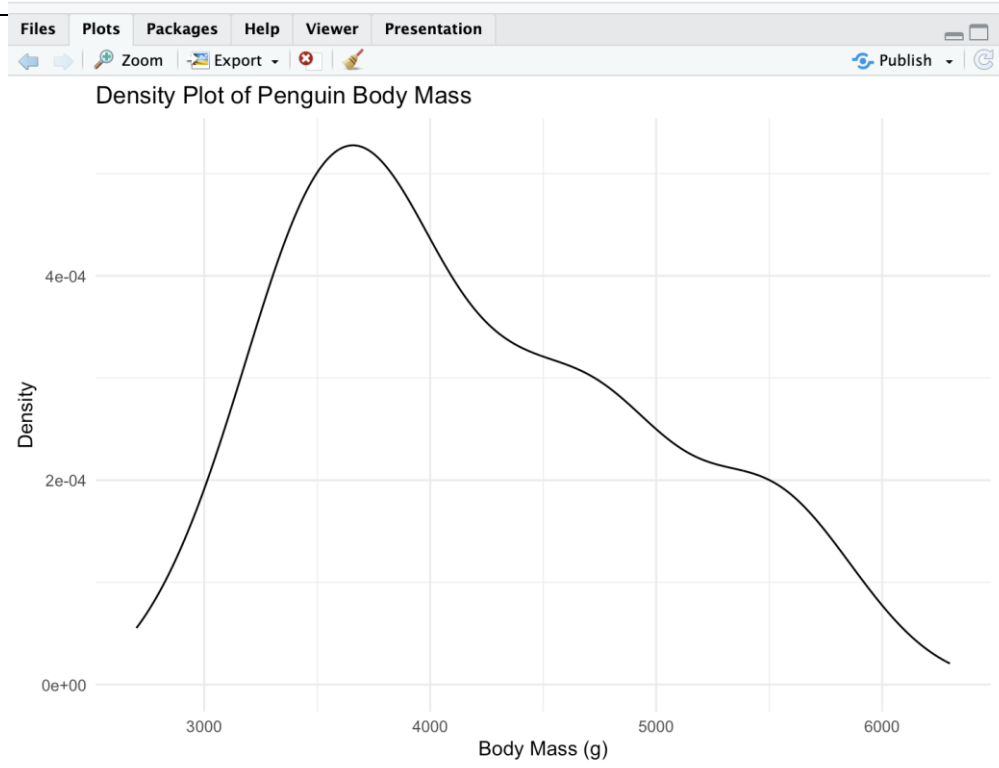
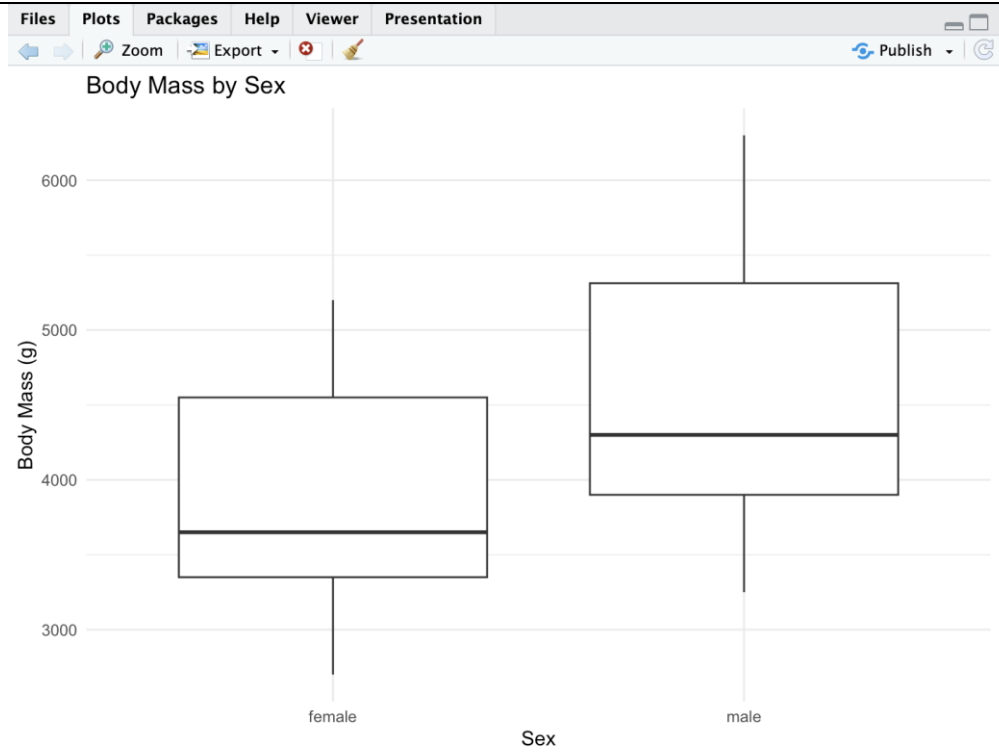
>
> cat("\n===== \n")

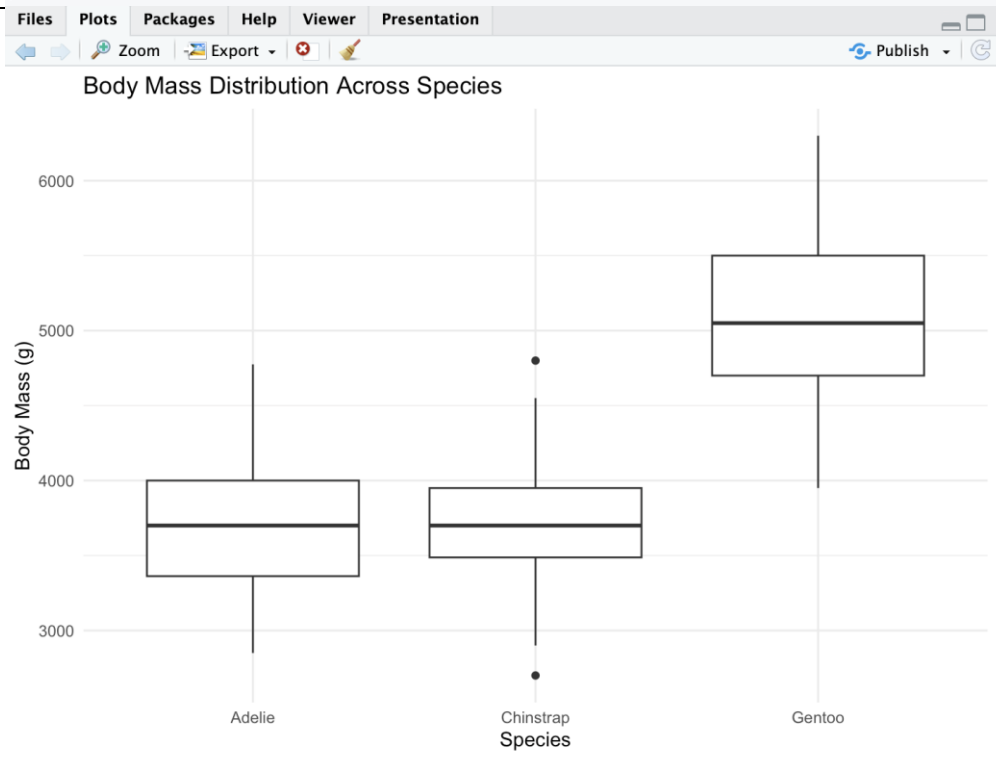
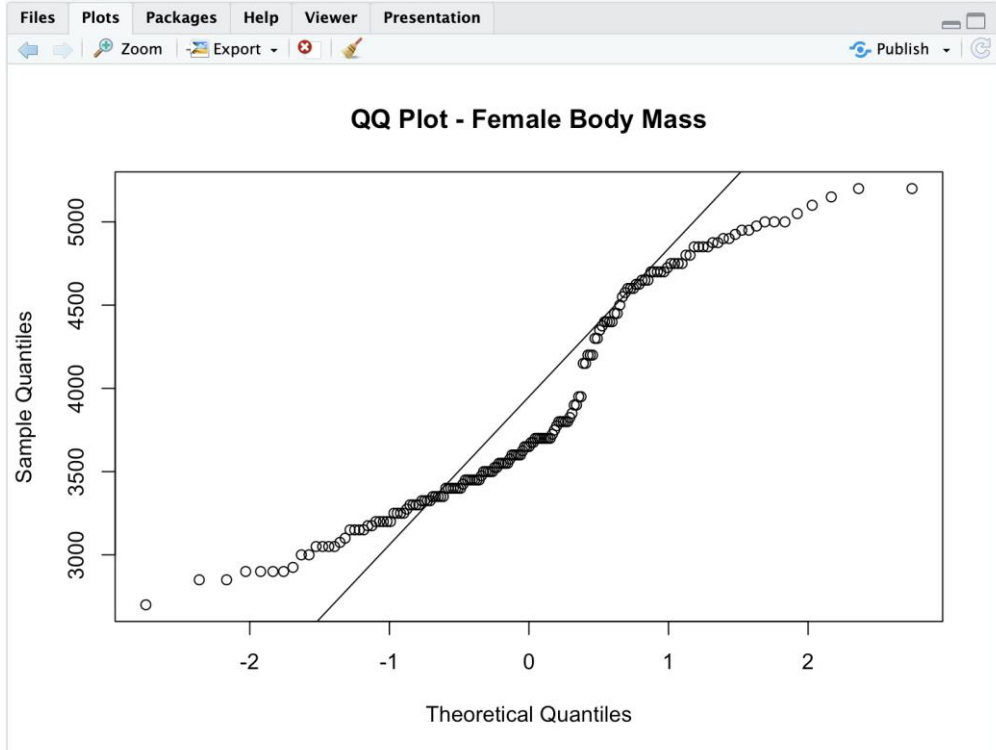
=====

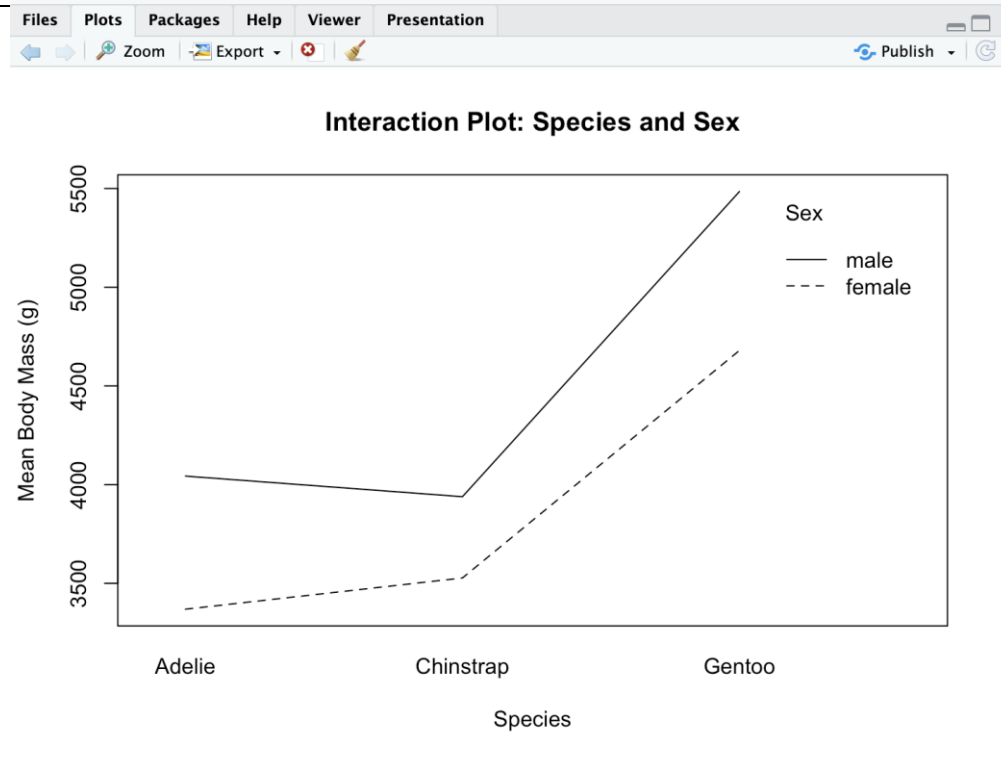
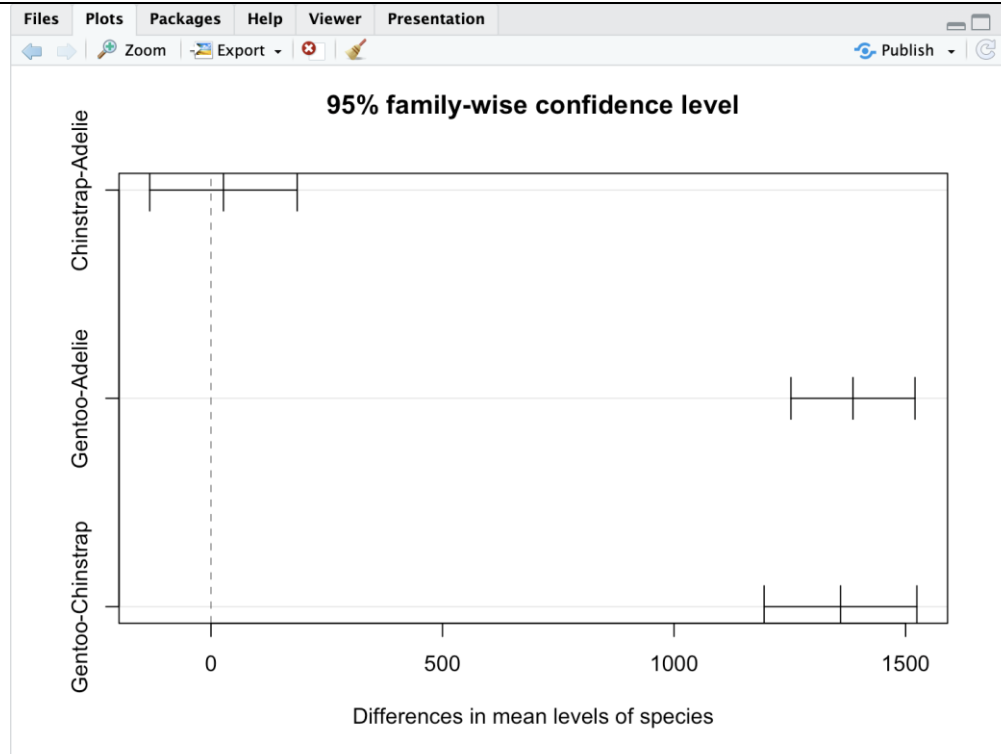
> |

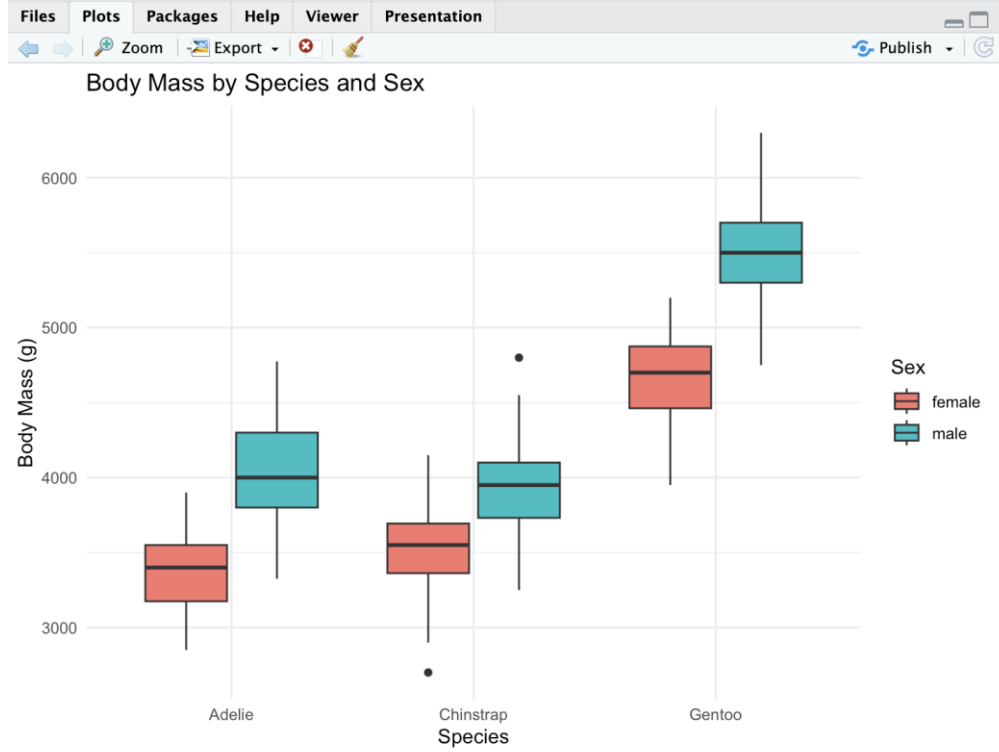
```

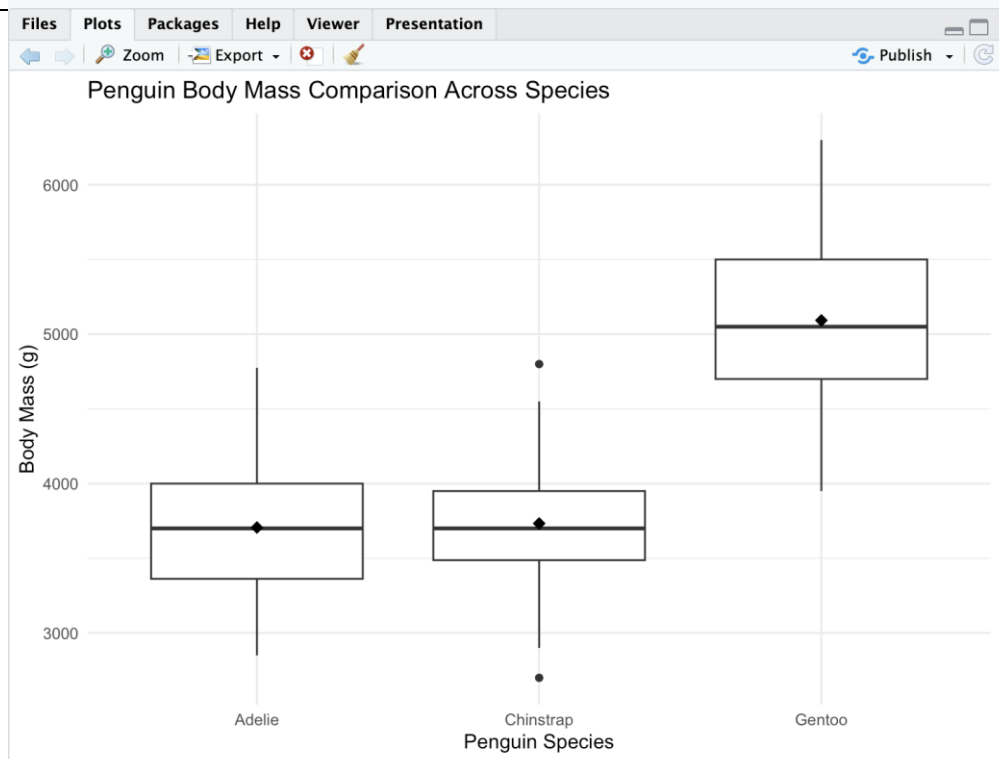
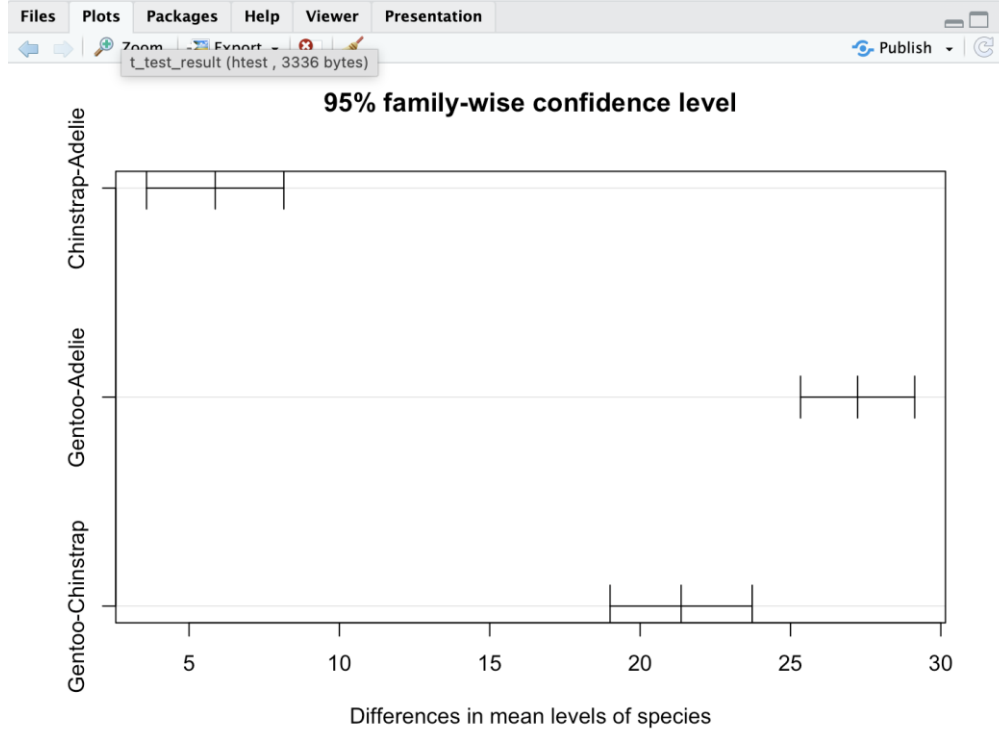












	Data	
	• anova_model	List of 13 
	• cohen_result	List of 8 
	• flipper_anova	List of 13 
	• flipper_data	342 obs. of 8 variables 
	• flipper_kruskal	List of 5 
	• flipper_stats	3 obs. of 6 variables 
	• flipper_tukey	List of 1 
	• kruskal_result	List of 5 
	• penguin_data	333 obs. of 8 variables 
	• penguins	344 obs. of 8 variables 
	• species_stats	3 obs. of 12 variables 
	• t_test_result	List of 10 
	• tukey_result	List of 1 
	• two_way_model	List of 13 
	Values	
	body_mass	int [1:333] 3750 3800 3250 3450 3650 3625 4675 3200 3800 4400 ...
	penguins_raw	<Promise>